

Spam email detection using BERT and GPT models

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Abstract

Spam email detection remains a critical challenge in cybersecurity, as malicious emails continue to evolve in sophistication. Traditional methods often rely on handcrafted features or shallow machine learning models, which struggle to capture the complex semantic and contextual patterns in modern spam emails. In this paper, we try to implement a spam email detection system using the GPT-2 model while using BERT as the base model to find out the potential performance improvement of the GPT model.

After comparing the performance of the two models, we found that the BERT and GPT-2 models generate similar state-of-the-art metrics, while for the GPT-2 model, training time increases a lot and the result improve tiny bit compared to the BERT model.

1 Introduction

With the rapid development of the Internet, email has become a fast and convenient method of communication that is widely used in our daily lives.

However, the popularity of email has also caused a problem that cannot be ignored: spam. Unscrupulous businessmen and criminals have exploited this communication medium to send spam with commercial advertisements and malicious or illegal files to users on a large scale (Kumar et al., 2020). Those spam emails not only occupy network resources and cause network congestion, but also affect people's normal work and life.

Therefore, how to quickly and accurately check whether an email is spam or not becomes a major topic worthy of research. It can protect people's property and privacy security and also improve people's work and life efficiency.

In this paper, we will try to solve the spam email detection problem using the latest transformer-based models, and using the TREC 2007 spam email dataset to train our model (Cormack, 2007).

Considering the growing use of pre-trained language in the NLP systems, in this experiment, we will adopt two distinct transformer-based models, BERT(encoder-only model) (Devlin et al., 2018a) and GPT-2(decoder-only model) (Radford et al., 2019) for this spam detection task and verify their real performance on our task. This comparative study will also try to find out if the generative pre-trained (GPT-2) model can provide any advantages or disadvantages in a spam detection task.

2 Theory

2.1 Transformer & Models Used

The transformer architecture used in the natural language processing (NLP) field is a revolutionary technology. By replacing traditional recurrent and convolutional layers with a mechanism based on self-attention (Vaswani et al., 2023), it enables the model can capture the contextual relationships more effectively than previous architectures.

2.2 BERT & GPT-2

The models we will use in this study are two popular transformer-based models, BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer). They are two distinct models with different component stacks, and they are designed with different approaches. BERT (Koroteev, 2021) is an encoder-only model based on a bidirectional deep learning model and can capture the information from both directions, making it suitable to understand language structure and can handle tasks like question answering, named entity recognition, etc. In contrast, GPT and its new versions (GPT-2, GPT-3, and GPT-4) are decoder-only models which focus on an auto-regressive approach (Zhao et al., 2025), and their tokens are based on a left-to-right manner, which makes it more suitable to generate creative content.

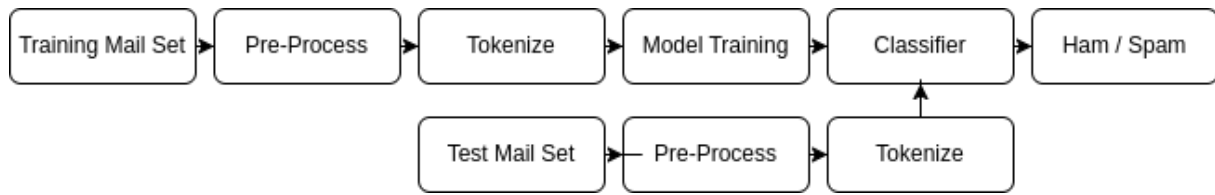


Figure 1: Spam email classification process

2.3 Evaluation Metrics

We will use the commonly used confusion matrix, accuracy, precision, recall, and F1-Score as evaluation indicators on the test set, which can provide clear and interpretable metrics for assessing classification performance. We will also compare train loss and test loss to find the most suitable epoch number.

3 Data

This paper uses the TREC07 spam corpus (Cormack, 2007) as the experimental dataset, which is downloaded from Kaggle, and it is a public dataset provided by the International Conference on Text Retrieval. The TREC07 dataset includes 29923 spam emails (56%) and 23745 ham emails (44%).

The original data columns are as follows:

- **label:** Email label, 1 for spam / 0 for ham.
- **origin:** Email content, containing all the text in an email.

And the sample data are listed below.

1(Spam). *Do you feel the pressure to perform and not rising to the occasion? Try via gr a—your anxiety will be a thing of the past and you will be back to your old self.*

0(Ham). *Hm, sounds like a homework problem to me. Maybe start by figuring out how to do it without R. What's the approach, and how would you calculate it? Then search R help for the possible keywords you came up with.*

4 Method

4.1 Data Preprocessing

The dataset we used does not suffer from significant class imbalance (56% vs 44%), so an additional dataset balance operation will not be executed, and the data pre-processing steps used in this paper will only include the following steps:

- Replace the abbreviations with their normal form.
- Remove all the stop words.
- Lemmatise the words with their normal form.

We will use functions provided by the NLTK library to remove the stop words and lemmatise the words. Meanwhile, we also use the regular expression library to replace the abbreviations with their normal form to make the data cleaner and easier to process. Then the data will be tokenised and converted into a format that can be used by the model. The processed data will be split into training, validation, and test sets using a 70%-15%-15% ratio.

4.2 Email spam text classification process

The Email spam text classification process is shown in Figure [1] below. The process includes the following steps:

- Preprocessing the data
- Tokenization
- Model building and training
- Spam email classification

Loss value will be calculated every epoch for both training and validation datasets, but accuracy, precision, recall, and F1-Score will be available only on the test dataset. With this in mind, we will use 30 epochs for each model.

4.3 BERT model

The BERT model used here is a pre-trained model google-bert/bert-base-uncased (Devlin et al., 2018b), which can be downloaded from the Hugging Face model hub. It has 110M parameters, and it contains 2 linear layers with 256 hidden units, dropout rate set to 0.2, and use RELU function as the activation function. The loss function used

in the BERT model is Binary Cross-Entropy Loss, and the optimiser used is the Adam optimiser with a default learning rate.

4.4 GPT model

GPT model, on the other hand, use the pre-trained model GPT-2 provided by OpenAI(Radford et al., 2019). However, the GPT model has a different input from the BERT model, so the split input data will be modified to fit the GPT model accordingly. Loss function used in the GPT model is Cross Entropy Loss instead of Binary Cross Entropy Loss, and optimizer used is Adam optimiser with default learning rate. GPT-2 model has 124M parameters, and it contains 2 linear layers with 256 hidden units, a dropout rate set to 0.2, and use RELU function as the activation function.

4.5 GPT-2 for Sequence Classification

Although GPT-2 was originally designed as a text generation language model, it can be effectively adapted to handle a supervised sequence classification task. The key idea behind this is to reuse the pre-trained GPT-2 transformer layer and attach a lightweight classification head on it, and it uses the last token to do the classification.

Given an input email token sequence, the GPT model computes a sequence of hidden representations through its self-attention layers. For sequence classification, the resulting representation is passed to a linear classification layer with two output labels, corresponding to the binary classification setting. In the case of binary spam detection, this corresponds to predicting whether an email belongs to the ham or spam class. We will use GPT2ForSequenceClassification class provided by Hugging Face to implement this feature(HuggingFace).

4.6 Evaluation

This article uses the following five values as evaluation indicators. 1. Confusion Matrix 2. Accuracy. 3. Precision. 4. Recall. 5. F1-score. We calculate the confusion matrix as the base result, and provide extended metrics based on this. Accuracy represents the spam detection rate, which is the accuracy of the model's classification results. Precision represents the accuracy of the classifier. Recall represents the spam detection rate, and F1 value is a combination of the precision and recall of the classifier, which is the harmonic average of the two, and it is a value between 0 and 1.

	Ham	Spam
Ham	3525	37
Spam	58	4431

Table 1: BERT Confusion Matrix

5 Results

5.1 Experiment Environment

The experiment in this paper is conducted on Google Colab(Google) with an NVIDIA GeForce A100 graphics card to minimise the training time. The main libraries used in this experiment includes nltk, pytorch, scikit-learn, transformers and matplotlib libraries. For the code related to this experiment, please refer to the following link.¹

5.2 Experiment Results

In this experiment, we employed the BERT model; its loss and accuracy are illustrated in Figure [2]. The plotted results indicate that the training loss consistently decreased from 0.12 to around 0.01. Meanwhile, the validation loss shows a decreasing trend at the beginning, from around 0.10 to around 0.04. However, after 13 epochs, an increase in the validation loss was observed, suggesting potential overfitting.

For the accuracy of validation data, validation accuracy increased from around 0.970 to 0.990, with some fluctuations after epoch 10. From the plot, we can say that the BERT model achieves a pretty good performance.

For the GPT-2 model, in Figure [3], we notice that the training loss decreases sharply from 0.14 to around 0.01 in 4 epochs, and then decreases slowly and approach 0 after epoch 15. However, we also notice that validation loss decreases at the beginning, but after the epoch 3, it begins to increase rapidly with some fluctuations. It is potential overfitting. For the accuracy of validation data, validation accuracy increases from 0.986 with a slow upward trend, but with several big fluctuations.

We list the final confusion matrix in Table[1] and Table[2], four values of the BERT and GPT2 models are also listed in Table [3] using the corresponding value of epoch number that minimises the validation loss.

¹Source Code:https://github.com/qyfly/text_mining_project

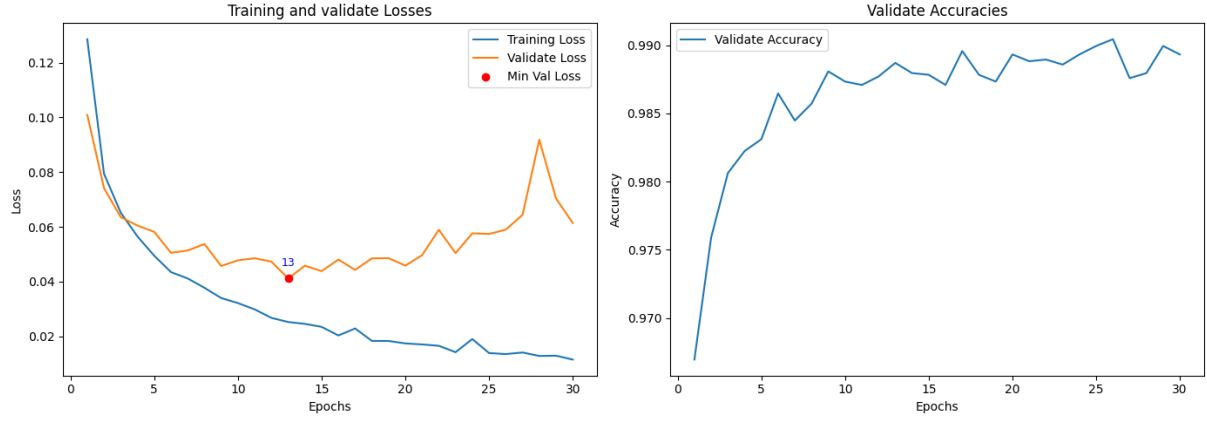


Figure 2: BERT Model Loss And Accuracy

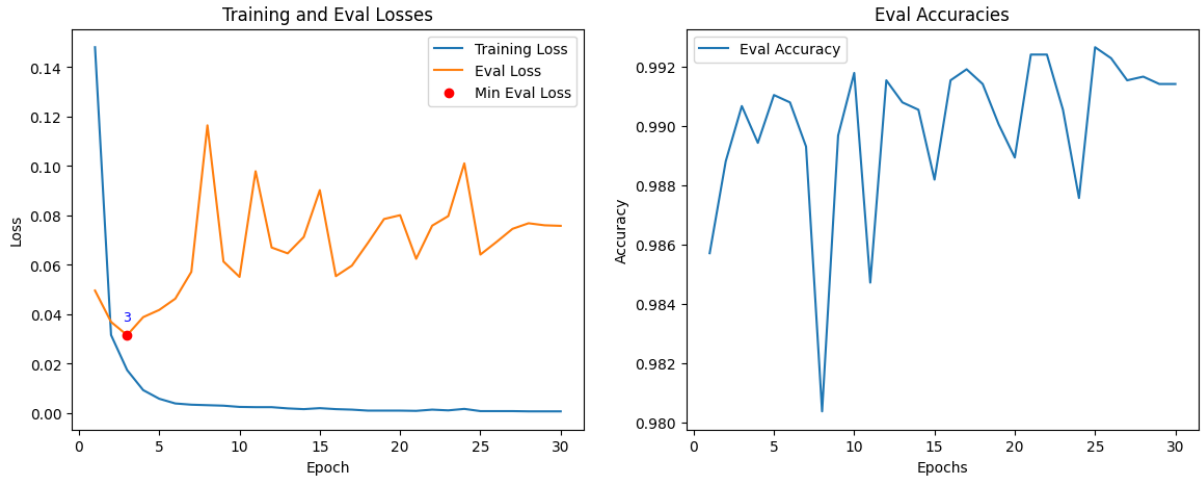


Figure 3: GPT-2 Model Loss And Accuracy

	Ham	Spam
Ham	3511	51
Spam	18	4471

Table 2: GPT-2 Confusion Matrix

Model	BERT	GPT-2
Epoch	13	3
Accuracy	0.9882	0.9914
Precision	0.9917	0.9887
Recall	0.9871	0.9960
F1-score	0.9894	0.9923

Table 3: BERT And GPT-2 Model Evaluation

6 Discussion

From the previous result, we notice that both models perform similarly. But we also have the following findings.

1. Well pre-trained Transformer-based models are good enough to handle easy NLP tasks like email ham/spam.

We notice that the metrics of both models are quite satisfactory in handling our task. Both models achieve very high results. This is mainly because spam detection is a relatively easy NLP task for modern LLMs and bert model. Because of the large parameters amount(110M and 124M), the pre-trained models have already seen massive amounts of similar text during pre-training. Fine-tuning on 37k labelled emails is more than sufficient for them to learn a near-perfect decision boundary.

Several papers also show similar results. Thae

states that in his experiment, he got 98.81% spam accuracy using BERT model(Sahmoud and Mikki, 2022) while Sergio states that using LLMs like GPT-2 he also got 97% accuracy(Rojas-Galeano, 2024).

Regarding the super high metrics we got, it may also be caused by the fact that the dataset we use is a pretty old public corpora(year 2007), which may already be well known by most of the LLMs and the latest BERT model.

2. Effective Epochs and Training Time

In our experience, even with 30 epochs, BERT can run much faster, in around 10 minutes, while for the GPT-2 model, the whole processing time will be around 30 minutes, even if we set the `per_device_train_batch_size` and `gradient_accumulation_steps` to a relative large number, which is 64 and 2, respectively.

In the GPT-2 experiment, we also found that after 3 epochs, the eval loss began to increase, which makes sense, because in most of the LLMs supervised training, we will only train the input 2-4 rounds, not 30 in our case, which leads to the overfitting issue in our GPT-2 experiment.

Regarding the limitations of this work, we can point out the following:

1. Limited Dataset

We only use the text data instead of HTML mail, which may contain more multimedia information, such as pictures and video, so the model we use may not be able to detect spam emails with multimedia information. Also, the dataset is pretty old compared to the latest models in Hugging Face; this setting pushes the metrics super high in our experiment.

2. Limited Models

We only compare the BERT and GPT models to detect spam emails, and there are many other models that can be used to detect spam emails, like DistilBERT (Sanh et al., 2019), RoBERTa (Liu et al., 2019), ALBERT (Lan et al., 2019), which may generate better performance than BERT.

About the future work and the limitations mentioned above, we need to use more advanced models to detect spam emails, and to identify those spam emails with multimedia information.

7 Conclusion

In this study, we aimed to compare the performance of BERT and GPT-2 in the task of spam email classification.

After training and evaluating both models, we have the following key findings.

- Bidirectional and Unidirectional Processing not matter when the base models are well pre-trained.

BERT checks both past and future tokens in a text so it can get richer contextual information. BERT's pretraining objectives made it suitable for classification tasks.

In contrast, GPT-2's left-to-right processing limited its ability to capture later token information, which let the model know less context information about the email.

However, when the base models are well trained, the performance(confusion matrix-related metrics) difference between models can be ignored, and they can get similar results.

- Training Time Variability Among Transformer-Based Models :

Even if those two models are Transformer-Based, the training time will be different, and we should choose a suitable model for a specific task instead of using a generalised model.

7.1 Extent to Which the Problem Was Solved

In addition to comparing the performance of BERT and GPT-2, we also gained some important insights from this study:

- Latest transformer-based model already knows old data well:

Our experiments indicate that modern transformer-based language models achieve near-saturation performance on old datasets. In tasks such as spam-ham classification on TREC 2007 dataset, these models consistently reach very high accuracy with minimal fine-tuning, suggesting that much of the relevant linguistic and semantic structure has already been captured during pre-training. This implies that improvements observed on such datasets may reflect prior knowledge encoded in the model. Consequently, commonly used legacy datasets may no longer provide sufficient challenge for the capabilities of contemporary transformer architectures. So we should always use the latest dataset for training, eval and testing, or gather the current

data by ourselves, to reflect the real-world spam email trend and then identify those spam emails.

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